QUANTITATIVE RESEARCH METHODS · FALL 2026 · TOPICS MISSING FROM YOUR NOTES

Research Methods: Missing Topics

The session-plan topics that your four class files do not cover, or cover only in part — explained from zero. 46 subtopics. Each starts with 'In plain words', then tables, formulas and fully worked examples. Read it together with your Research Statistics Summary.

★★★★★ must know, asked often ★★★★☆ high ★★★☆☆ medium ★★☆☆☆ low ★☆☆☆☆ skim

‡ rule or number that changes often; check workbook edition

Ch	Chapter	Marks	Priority	Subtopics	5★
1	Introduction to Research & Problem Formulation	Module 1 · Class 1	★★★★☆	5	4
2	Research Design & Types of Research	Module 1 · Class 2	★★★★★	7	5
3	Methods of Data Collection: Primary & Secondary Data	Module 2 · Classes 3–4	★★★★★	6	3
4	Measurement & Scaling Techniques	Module 2 · Classes 5–6	★★★★★	5	4
5	Questionnaire Design	Module 2 · Classes 5–6	★★★★★	5	4
6	Point & Interval Estimates (Confidence Intervals)	Module 3 · Classes 9–10	★★★★★	7	7
7	t-Tests in Full, Power of a Test & Degrees of Freedom	Module 3 · Classes 7–13	★★★★★	5	5
8	Report Writing Mechanics	Module 4 · Classes 22–23	★★★★★	5	2
9	Coverage Map: Session Plan vs Your Notes	—	★★★★☆	1	0

5★ list (34) — read these first

Ch 1 · The research process (step by step) · Problem formulation: management problem vs research problem · How to define the problem · Developing the approach: research questions & hypotheses

Ch 2 · What a research design is · Types of research (classifications) · Exploratory, descriptive and causal (conclusive) research · Descriptive designs: cross-sectional vs longitudinal · Causal research: experiments, variables and validity

Ch 3 · Primary vs secondary data · Sources and evaluation of secondary data · Survey methods

Ch 4 · Measurement and scaling: basic ideas · Comparative scaling techniques · Non-comparative scaling: continuous and itemised rating scales · Reliability and validity

Ch 5 · The 10 steps of questionnaire design (Malhotra) · Question structure · Question wording rules · Order, layout and pretesting

Ch 6 · Point vs interval estimates · General form of a confidence interval · CI for a mean: σ known or large sample (z) · CI for a mean: σ unknown, small sample (t) · CI for a proportion · CIs for differences (analysis of differences) · Sample size for estimation

Ch 7 · Two independent samples t-test (pooled variance) · Paired (dependent) t-test: worked · Which mean test? (summary) · Power of a test · Degrees of freedom

Ch 8 · Structure and components of a research report · Guidelines for writing

4★ list (12)

Ch 1 · What research is

Ch 2 · Experimental designs · Saunders' research onion

Ch 3 · Qualitative primary methods · Observation and experiments · Sampling: steps and sample size (supplement)

Ch 4 · Multi-item scales and measurement error

Ch 5 · Why questionnaire design matters

Ch 8 · Purpose and types of research reports · Tables and graphs · Oral presentation and follow-up

Ch 9 · Where to study each session-plan topic

Sources: the standard texts on your session plan — Malhotra (Marketing Research), Zikmund (Business Research Methods), Saunders, Lewis & Thornhill (Research Methods for Business Students), Kothari (Research Methodology), Levin & Rubin (Statistics for Management), Chawla & Sondhi (Research Methodology). Worked examples are written for this summary; check the numbers your professor uses in class.

01 Introduction to Research & Problem Formulation

≈Module 1 · Class 1 marks · 5 subtopics · chapter priority ★★★★☆

What research is ★★★★☆

In plain words: research is a careful, organised search for an answer to a question. Business research collects and analyses information so that managers can make better decisions.

Feature of good research	Meaning
Systematic	Follows clear, planned steps
Logical	Reasoning (induction/deduction) links evidence to conclusions
Empirical	Based on real data you observe or measure
Replicable	Others can repeat it and check the result
Objective	Free from the researcher's personal bias
Purpose-driven	Clear objective; limits stated honestly

Kind of reasoning	How it works	Example
Deductive	Theory → hypothesis → data → accept/reject	Theory says price cuts raise sales → test in stores
Inductive	Data → patterns → theory	Interviews show customers value delivery speed → new theory
Abductive	Move back and forth between data and theory	Surprising result → best explanation → test again

The research process (step by step) ★★★★★

In plain words: every study follows the same order — define the problem, plan, collect data, analyse, report. A mistake early (wrong problem) spoils everything later.

Kothari (11 steps)	Malhotra (6 steps)
1. Formulate the research problem	1. Define the problem
2. Extensive literature survey	2. Develop an approach (theory, models, research questions, hypotheses)
3. Develop working hypotheses	
4. Prepare the research design	3. Formulate the research design
5. Determine the sample design	
6. Collect the data	4. Fieldwork / data collection
7. Execute the project	
8. Analyse the data	5. Data preparation and analysis
9. Test hypotheses	
10. Generalise and interpret	
11. Prepare the report	6. Report preparation and presentation

The process is not strictly linear — results can send you back to an earlier step

Literature review: find what is already known, the gaps, and theories and measures you can reuse

Problem formulation: management problem vs research problem ★★★★★

In plain words: the manager asks 'what should I **do**?' The researcher turns it into 'what do I need to **know**?' Getting this translation right is the most important step.

Management decision problem	Research problem
Asks what the decision maker must do	Asks what information is needed and how to get it
Action-oriented	Information-oriented
Focus on symptoms	Focus on underlying causes
Should a new product be launched?	Find consumer preferences and purchase intention for the new product
Should the advertising campaign be changed?	Measure how effective the current campaign is
Should the price be raised?	Estimate price elasticity of demand and the effect of price changes on sales and profit

Symptom vs problem: falling sales is a **symptom**; the problem may be a new competitor, poor service or wrong price

How to define the problem ★★★★★

Task	Purpose
Discussion with decision makers	Learn the objectives, history, alternatives and what decision the research must support (problem audit)
Interviews with industry experts	Expert knowledge, quick and cheap
Secondary data analysis	Facts already published — check before collecting new data
Qualitative research	Small exploratory studies (focus groups, depth interviews) to understand the issue

Environmental context to study	Examples
Past information and forecasts	Sales, market share, industry trends
Resources and constraints	Budget, time, skills
Objectives	Of the organisation and the decision maker
Buyer behaviour	Who buys, why, how
Legal, economic environment	Regulation, income, interest rates
Marketing and technology skills	What the firm can actually do

```
Problem statement = broad statement (general problem) + specific components (exact questions)
Broad: 'Improve the store's image'
Components: 1. What criteria do shoppers use to choose a store?
2. How do shoppers rate our store and competitors on these criteria?
3. Which customer groups visit us?
```

TRAP Common errors: problem too **BROAD** (no clear guide) or too **NARROW** (misses options). Use a broad statement plus specific components.

Developing the approach: research questions & hypotheses ★★★★★

Component	Meaning
Objective / theoretical framework	Theory from the literature that guides what to measure
Analytical model	Variables and their links: verbal, graphical (diagram) or mathematical (equation)
Research questions (RQs)	Refined questions about the specific components
Hypotheses (H)	Testable, unproven statements that answer an RQ
Information needed	Exact list of data required

```
RQ: Do loyal customers of the store differ in income from non-loyal customers?
H1: Loyal customers have higher average income than non-loyal customers
H0: There is no difference in average income
```

A good research problem / hypothesis is...	...because
Clear and specific	Everyone understands it the same way
Testable / researchable	Data can answer it
Relevant and significant	Supports a real decision
Feasible	Time, money, data and skills exist
Ethical	No harm to participants

02 Research Design & Types of Research

≈Module 1 · Class 2 marks · 7 subtopics · chapter priority ★★★★★

What a research design is ★★★★★

In plain words: a research design is the blueprint for the study — what data, from whom, how, when, and how it will be analysed. Your notes list designs; here are the types you are examined on.

Parts: information needed · exploratory, descriptive or causal phase · measurement and scaling · questionnaire · sampling and sample size · data analysis plan

Good design = answers the research questions with the least error at an acceptable cost

Types of research (classifications) ★★★★★

Basis	Types	Meaning
Purpose	Basic (pure / fundamental)	Builds general knowledge and theory
	Applied	Solves a specific practical problem
Method	Descriptive	Describes the state of affairs as it is (no control of variables)
	Analytical	Uses existing facts to critically evaluate
Data	Quantitative	Numbers, measurement, statistics
	Qualitative	Words, meanings, motives
Basis of knowledge	Conceptual	Abstract ideas and theory
	Empirical	Observation and experiment; tests hypotheses with data
Time	Cross-sectional	Data at one point in time
	Longitudinal	Same units measured repeatedly over time

Exploratory, descriptive and causal (conclusive) research ★★★★★

	Exploratory	Descriptive	Causal
Objective	Gain ideas and insights; define the problem	Describe market traits or functions	Find cause-and-effect
Characteristics	Flexible, unstructured	Pre-planned, structured; clear hypotheses	Manipulate one or more independent variables; control others
Sample	Small, non-representative	Large, representative	Matched groups
Methods	Expert surveys, pilot studies, secondary data, qualitative research (focus groups, depth interviews)	Surveys, panels, observation, secondary data	Experiments
Findings	Tentative — followed by more research	Conclusive	Conclusive

Descriptive and causal research together = **conclusive** research

Saunders also uses **explanatory** research = studies a situation to explain links between variables (similar to causal)

Descriptive answers the **6 Ws**: who, what, when, where, why, way

> **TRAP** Exploratory findings are **NOT** used directly for decisions — they generate hypotheses.

Descriptive designs: cross-sectional vs longitudinal ★★★★★

Design	How	Strength	Weakness
Single cross-sectional	One sample, measured once	Representative snapshot	No change over time
Multiple cross-sectional	Two or more samples, each measured once (cohort analysis)	Compare groups / generations	Different people each time
Longitudinal (panel)	Same sample measured repeatedly	Shows change in the same units	Panel attrition, panel conditioning, costly

Cohort analysis: follow groups born in the same period across surveys (e.g. soft-drink use by age group over decades)

Causal research: experiments, variables and validity ★★★★★

In plain words: to prove X causes Y, change X yourself, keep other things fixed, and see if Y changes.

Condition for causality	Meaning
Concomitant variation	X and Y move together as the hypothesis predicts
Time order	X happens before (or at the same time as) Y
Elimination of other causes	No other factor explains the change

Term	Meaning
Independent variable (treatment)	The variable you change (price, ad)
Dependent variable	The outcome you measure (sales)
Extraneous variable	Other factors that can affect the outcome
Test units	People, stores or firms that receive treatments
Internal validity	Did the treatment really cause the effect?
External validity	Can results be generalised beyond the experiment?

Threat to internal validity	Meaning
History	Outside events during the experiment
Maturation	Test units change with time (tired, older)
Testing (main / interactive)	Earlier measurement affects later answers
Instrumentation	Measuring tool or observers change
Statistical regression	Extreme scorers move toward the mean
Selection bias	Groups differ at the start
Mortality	Test units drop out

Control by: **randomisation**, **matching**, statistical control (ANCOVA), design control

Experimental designs ★★★★☆

```
Notation: X = treatment, O = observation (measurement), R = random assignment, EG = experimental group, CG = control group
```

Type	Design	Symbol
Pre-experimental (no randomisation)	One-shot case study	X O_1
	One-group pretest–posttest	O_1 X O_2
	Static group	EG: X O_1 / CG: O_2
True experimental (randomised)	Pretest–posttest control group	EG: R O_1 X O_2 / CG: R O_3 O_4 → effect = (O_2 − O_1) − (O_4 − O_3)
	Posttest-only control group	EG: R X O_1 / CG: R O_2 → effect = O_1 − O_2
	Solomon four-group	Adds two posttest-only groups to measure testing effects
Quasi-experimental	Time series	O_1 O_2 O_3 X O_4 O_5 O_6
	Multiple time series	Time series plus a control group
Statistical	Randomised block	Block on one extraneous variable (e.g. store size)
	Latin square	Control two extraneous variables
	Factorial	Two or more treatments and their interaction (link to two-way ANOVA)

Lab experiment: high internal validity · Field experiment (real setting): high external validity

Test marketing = a field experiment for a new product

Saunders' research onion ★★★★☆

Layer (outside → in)	Choices
Philosophy	Positivism, critical realism, interpretivism, postmodernism, pragmatism
Approach to theory	Deductive, inductive, abductive
Methodological choice	Mono method (quantitative or qualitative), multi-method, mixed methods
Strategy	Experiment, survey, archival, case study, ethnography, action research, grounded theory, narrative inquiry
Time horizon	Cross-sectional, longitudinal
Techniques and procedures	Data collection and analysis

03 Methods of Data Collection: Primary & Secondary Data

≈Module 2 · Classes 3–4 marks · 6 subtopics · chapter priority ★★★★★

Primary vs secondary data ★★★★★

In plain words: primary data you collect yourself for this study. **Secondary** data someone already collected for another purpose. Always check secondary data first — it is quicker and cheaper.

	Primary data	Secondary data
Collected for	The problem at hand	Other purposes
Process	Very involved	Rapid and easy
Cost	High	Relatively low
Time	Long	Short
Fit to problem	Exact	May not fit (units, dates, definitions)
Control over quality	Full	None

Sources and evaluation of secondary data ★★★★★

Source	Examples
Internal	Sales records, invoices, customer databases, accounting data, CRM, loyalty cards
External: published	Government (Census, RBI, NSSO/MoSPI), industry reports, journals, books, newspapers
External: computerised databases	Online, offline, bibliographic, numeric, full-text, directory
External: syndicated services	Paid data sold to many clients: consumer panels, retail audits (Nielsen), media audience data

Criterion to judge secondary data	Question to ask
Specifications / methodology	How was it collected — sample, response rate, fieldwork?
Error / accuracy	Is it accurate? Compare sources
Currency	When was it collected? Is it out of date?
Objective	Why was it collected? (bias risk)
Nature	Definitions, units, categories — do they fit?
Dependability	Is the source expert, credible, reputable?

TRAP Secondary data is a starting point, rarely a complete answer — definitions and dates often do not match your problem.

Qualitative primary methods ★★★★☆

Method	How	Use / limits
Focus group	8–12 people, a moderator, 1–2 hours	Ideas, reactions, new concepts; not representative
Depth interview	One-to-one, unstructured, 30 min–1 hr+	Sensitive topics, experts; costly, depends on interviewer
Projective techniques	Indirect tasks that reveal hidden motives	Word association, sentence completion, picture response (TAT), cartoon tests, role playing, third-person technique

Direct approach (purpose clear: focus groups, depth interviews) vs indirect (purpose disguised: projective techniques)

Survey methods ★★★★★

In plain words: a survey asks many respondents structured questions. The mode (how you ask) affects cost, speed, response rate and quality.

Mode	Strengths	Weaknesses
Personal (in-home, mall intercept, CAPI)	Complex questions, visual aids, high response	Costly, interviewer bias
Telephone (CATI)	Fast, wide reach, supervision	Short only, no visuals, low response now
Mail	Cheap, no interviewer bias, anonymous	Low response, slow, no control over who answers
Electronic (email, web, mobile)	Very cheap, fast, multimedia, automatic data entry	Sample may not represent population; low response; security

Kothari: questionnaire vs schedule	Questionnaire	Schedule
Who fills it	Respondent (by post/online)	Enumerator, face to face
Cost	Cheap	Costly (field staff)
Non-response	High	Low
Respondent identity	Not always known	Known
Needs literate respondents	Yes	No

Observation and experiments ★★★★☆

Observation type	Meaning
Structured vs unstructured	Pre-set checklist vs record everything relevant
Disguised vs undisguised	Respondents unaware vs aware
Natural vs contrived	Real setting vs artificial setting
Personal	Human observer
Mechanical	Cameras, scanners, eye-tracking, people meters, web analytics
Audit	Count stock / pantry items
Content analysis	Analyse communication content (ads, posts)
Trace analysis	Physical traces (waste, browser history)

Observation: records actual behaviour, no recall bias — but cannot see motives or attitudes

Experiments: the only method that shows cause and effect (see Chapter 2)

Sampling: steps and sample size (supplement) ★★★★☆

Your notes cover sampling methods. These are the design steps around them.

```
1. Define the target population (elements, units, extent, time)
2. Determine the sampling frame (list of the population)
3. Select the sampling technique (probability or non-probability)
4. Determine the sample size
5. Execute the sampling process
```

Sample size depends on: importance of the decision, nature of research, number of variables, analysis planned, incidence and completion rates, resources

Formulas for sample size are in Chapter 6 (estimation)

Errors: **sampling error** (sample ≠ population) vs **non-sampling error** (wrong frame, non-response, measurement, data entry) — non-sampling errors are often larger

04 Measurement & Scaling Techniques

≈Module 2 · Classes 5–6 marks · 5 subtopics · chapter priority ★★★★★

Measurement and scaling: basic ideas ★★★★★

In plain words: measurement = giving numbers to characteristics by a rule (1 = male, 2 = female). **Scaling** = placing objects on a continuum (very unhappy … very happy). Your notes cover the four scale levels (nominal, ordinal, interval, ratio); this chapter covers the scaling **techniques**.

Scale level (recap)	What you can say	Allowed statistics
Nominal	Categories only	Count, mode, chi-square
Ordinal	Order	Median, percentiles, rank correlation
Interval	Equal gaps, no true zero	Mean, SD, correlation, t, ANOVA
Ratio	True zero	All statistics, incl. ratios

Comparative scaling techniques ★★★★★

In plain words: respondents compare objects **directly** with each other. Results are ordinal (relative), not absolute.

Technique	How	Example / note
Paired comparison	Choose one of two objects, for every pair	n brands → n(n − 1)/2 pairs: 5 brands = 10 pairs; transitivity check
Rank order	Rank all objects together	Rank 5 shampoos 1–5; quick, realistic; n − 1 decisions
Constant sum	Share a fixed total (e.g. 100 points) across items	Brand attributes: price 40, quality 35, design 25; gives ratio-like data
Q-sort	Sort many statements into piles by criteria	Used for large numbers of items (60–140)

Pros: easy to understand, small differences show up · Cons: ordinal only; results can't be generalised beyond the objects compared

Non-comparative scaling: continuous and itemised rating scales ★★★★★

In plain words: each object is rated **on its own**, not against others.

Scale	How	Example
Continuous (graphic) rating	Mark a point on a line	Worst ———|——— Best (score 0–100)
Likert (summated)	Agree/disagree with statements, usually 5 points	1 Strongly disagree … 5 Strongly agree; sum or average items
Semantic differential	7-point scale between two opposite adjectives	Cold 1 2 3 4 5 6 7 Warm; shows a profile of a brand
Stapel	Unipolar −5 to +5 with one adjective in the middle, no zero	+5 … +1 [High quality] −1 … −5; good for phone surveys

Scale decision	Options
Number of categories	5–9 usual; more = finer but harder
Balanced vs unbalanced	Equal favourable/unfavourable categories or not
Odd vs even	Odd has a neutral midpoint
Forced vs non-forced	Non-forced adds 'no opinion'
Verbal description	Label all, some or only end points
Physical form	Horizontal/vertical, numbers, faces

TRAP Likert data are strictly ordinal, but are usually **TREATED AS INTERVAL** (means, t-tests) in practice.

Multi-item scales and measurement error ★★★★☆

A construct (e.g. customer satisfaction) is measured with several items, then summed: develop theory → generate items → reduce items (expert judgement, pretest) → check reliability and validity → final scale

```
Observed score X_O = X_T (true score) + X_S (systematic error) + X_R (random error)
Reliability: low X_R ; Validity: low X_S and X_R  (X_O = X_T)
```

Reliability and validity ★★★★★

In plain words: reliable = gives the same result each time. **Valid** = measures what it should. A bathroom scale that always shows 2 kg too much is reliable but not valid.

Reliability type	How it is checked
Test–retest	Same scale, same people, two times; correlate
Alternative forms	Two equivalent versions; correlate
Internal consistency: split-half	Split items in two halves; correlate
Internal consistency: **Cronbach's alpha**	Average of all split-half coefficients; 0–1; **≥ 0.6** acceptable (≥ 0.7 often required)

Validity type	Meaning
Content (face) validity	Experts judge whether items cover the construct
Criterion validity	Agrees with a criterion: **concurrent** (same time) or **predictive** (future)
Construct validity	Measures the theoretical construct: **convergent** (correlates with related measures), **discriminant** (not with unrelated), **nomological** (fits theory)

Valid ⇒ reliable · Reliable does **not** ⇒ valid · Unreliable ⇒ not valid

Also: **generalisability** — can results extend beyond this sample and setting?

05 Questionnaire Design

≈Module 2 · Classes 5–6 marks · 5 subtopics · chapter priority ★★★★★

Why questionnaire design matters ★★★★☆

In plain words: a questionnaire is a fixed set of questions to get information from respondents. A badly worded question gives wrong data no matter how good the sample or analysis.

Objectives: turn information needs into questions people **can** and **will** answer · motivate respondents to finish · minimise response error

The 10 steps of questionnaire design (Malhotra) ★★★★★

```
1. Specify the information needed
2. Specify the type of interviewing method
3. Decide the content of individual questions
4. Design questions to overcome inability and
unwillingness to answer
5. Decide the question structure
6. Decide the question wording
7. Arrange the questions in proper order
8. Identify the form and layout
9. Reproduce the questionnaire
10. Pretest
```

Step 3: is the question necessary? Do you need several questions instead of one? (avoid **double-barrelled** questions)

Step 4 inability: is the respondent informed? can they remember? can they say it? — use filter questions, aided recall

Step 4 unwillingness: reduce effort, explain purpose, make the context fit; sensitive topics last, use third-person or counter-biasing statements, ranges (income bands)

Question structure ★★★★★

Type	Description	Pros / cons
Unstructured (open-ended)	Respondent answers in own words	Rich insight; hard to code, interviewer bias
Multiple choice	Choose from given answers	Easy to analyse; need full, exclusive options; order bias → rotate
Dichotomous	Two answers (yes/no), maybe a neutral	Simple; forces a choice; wording strongly affects result
Scales	Likert, semantic differential, etc.	Measure attitudes (see Chapter 4)

Question wording rules ★★★★★

Rule	Bad	Better
Define the issue (who, what, when, where, why, way)	Which brand of shampoo do you use?	Which brand(s) of shampoo have you personally used at home in the last month?
Use ordinary words	Is distribution adequate?	Can you find the product easily in shops?
Use unambiguous words	Do you shop often?	How many times did you shop in the last 4 weeks?
Avoid leading questions	Do you agree that patriotic Indians should buy Indian cars?	Which car brand would you buy?
Avoid implicit alternatives	Do you like to fly when travelling short distances?	Do you like to fly or drive on short trips?
Avoid implicit assumptions	Are you for a balanced budget?	...if it means higher taxes?
Avoid generalisations and estimates	What is your annual household spending on groceries?	How much do you spend on groceries in a week?
No double-barrelled questions	Is the food tasty and cheap?	Two separate questions
Use positive and negative statements	All items worded positively	Mix to reduce acquiescence bias

Order, layout and pretesting ★★★★★

Order rule	Reason
Opening questions simple and interesting	Build trust and interest
Qualifying (screening) questions first	Check the respondent belongs to the target
Basic information → classification → identification	Most important data first; personal details (income, name) last
Difficult or sensitive questions later	Rapport first
Funnel approach: general → specific	Avoids earlier questions biasing later ones
Logical order; branching (skip) questions	Clear flow; route people correctly

Layout: divide into parts, number questions, pre-code answers, keep it uncluttered; booklet form, clear instructions

Pretest on 15–30 people like the target respondents, with the same interview method; revise and pretest again

Checklist of errors: leading, double-barrelled, ambiguous, too long, missing options, sensitive questions too early

06 Point & Interval Estimates (Confidence Intervals)

≈Module 3 · Classes 9–10 marks · 7 subtopics · chapter priority ★★★★★

Point vs interval estimates ★★★★★

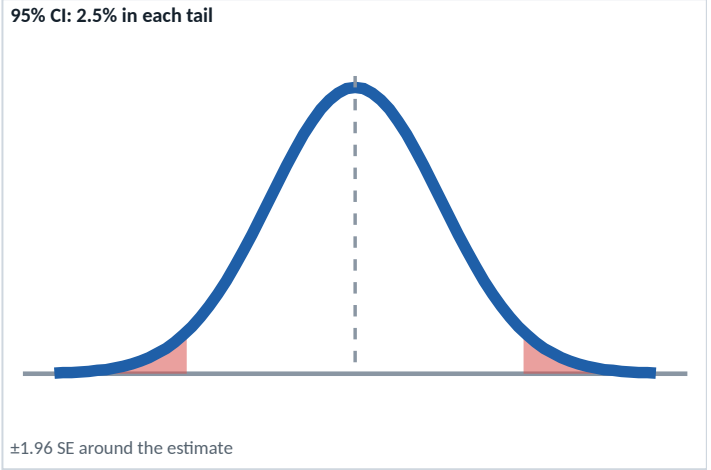
In plain words: a **point estimate** is one number used as a best guess for a population value (sample mean 50 → guess μ = 50). An **interval estimate** gives a range that probably contains the true value, with a stated **confidence level** (e.g. 95%).

Population parameter	Point estimator	Standard error (SE)
Mean μ	Sample mean $\bar{x}$	$\sigma/\sqrt{n}$ (or $s/\sqrt{n}$)
Proportion p	Sample proportion $\hat{p} = x/n$	$\sqrt{(\hat{p}\hat{q}/n)}$, $\hat{q} = 1 - \hat{p}$
Difference $\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	$\sqrt{(s_1^2/n_1 + s_2^2/n_2)}$
Difference $p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$	$\sqrt{(\hat{p}_1\hat{q}_1/n_1 + \hat{p}_2\hat{q}_2/n_2)}$
Variance σ^2	s^2 (divide by n − 1)	—

Quality of a good estimator	Meaning
Unbiased	Its average over all samples equals the parameter ($E(\bar{x}) = \mu$)
Efficient	Smallest standard error among unbiased estimators
Consistent	Gets closer to the parameter as n grows
Sufficient	Uses all the information in the sample

General form of a confidence interval ★★★★★

```
CI = point estimate ± (critical value × standard error)
Margin of error E = critical value × SE
z values: 90% → 1.645 ; 95% → 1.96 ; 99% → 2.576
```



Meaning of 95%: if we took many samples and built a CI from each, about 95% of them would contain the true μ

Wider interval when: higher confidence, smaller n, bigger σ

Link to tests: if a two-tailed 5% test value lies **outside** the 95% CI → reject H_0

TRAP Do not say 'there is a 95% probability that μ is in this interval' — μ is fixed; the interval is what varies.

CI for a mean: σ known or large sample (z) ★★★★★

```
x̄ ± z × σ/√n   (n ≥ 30: use s for σ)
n = 36, x̄ = 50, σ = 12, 95%
SE = 12/√36 = 2 ; E = 1.96 × 2 = 3.92
CI = 50 ± 3.92 = (46.08, 53.92)
Finite population (n/N > 5%): multiply SE by √[(N - n)/(N - 1)]
```

CI for a mean: σ unknown, small sample (t) ★★★★★

```
x̄ ± t(α/2, n - 1) × s/√n
n = 16, x̄ = 72, s = 8, 95% → df 15, t = 2.131
SE = 8/4 = 2 ; E = 2.131 × 2 = 4.262
CI = 72 ± 4.26 = (67.74, 76.26)
```

df	t one-tail 5% (two-tail 10%)	t two-tail 5%
5	2.015	2.571
10	1.812	2.228
15	1.753	2.131
20	1.725	2.086
30	1.697	2.042
∞ (= z)	1.645	1.960

t has fatter tails than z → wider interval; t → z as n grows

Assumes the population is roughly normal

CI for a proportion ★★★★★

```
p̂ ± z × √(p̂q̂/n)   (need np̂ and nq̂ ≥ 5)
120 of 400 customers prefer brand A → p̂ = 0.30
SE = √(0.30 × 0.70 / 400) = 0.0229 ; E = 1.96 × 0.0229 = 0.0449
CI = 0.30 ± 0.045 = (0.255, 0.345) → 25.5% to 34.5%
```

CIs for differences (analysis of differences) ★★★★★

In plain words: to compare two groups, build a CI for the **difference**. If the interval does not include 0, the groups really differ.

```
Means, large samples: (x̄₁ - x̄₂) ± z √(s₁²/n₁ + s₂²/n₂)
x̄₁ 85, s₁ 10, n₁ 50 ; x̄₂ 80, s₂ 12, n₂ 60
SE = √(100/50 + 144/60) = √4.4 = 2.098 ; E = 1.96 × 2.098 = 4.11
CI = 5 ± 4.11 = (0.89, 9.11) → excludes 0 → significant difference
```

```
Means, small samples (pooled): (x̄₁ - x̄₂) ± t(n₁ + n₂ - 2) × s_p √(1/n₁ + 1/n₂)
Proportions: (p̂₁ - p̂₂) ± z √(p̂₁q̂₁/n₁ + p̂₂q̂₂/n₂)
City A 80/200 = 0.40, city B 60/200 = 0.30
SE = √(0.24/200 + 0.21/200) = 0.0474 ; E = 0.093
CI = 0.10 ± 0.093 = (0.007, 0.193) → excludes 0 → A higher
```

Paired data: CI = $\bar{D} \pm t(n-1) \times s_D/\sqrt{n}$

Sample size for estimation ★★★★★

```
Mean: n = (z σ / E)²
σ = 12, E = 2, 95% → (1.96 × 12 / 2)² = 138.3 → 139 (always round up)
Proportion: n = z² p q / E²
No prior p → use p = 0.5 (largest n)
E = 0.03, 95% → 1.96² × 0.25 / 0.0009 = 1,067.1 → 1,068
```

Halving E needs four times the sample

Higher confidence or larger σ → larger n

07 t-Tests in Full, Power of a Test & Degrees of Freedom

≈Module 3 · Classes 7–13 marks · 5 subtopics · chapter priority ★★★★★

Two independent samples t-test (pooled variance) ★★★★★

In plain words: compare the means of two separate small groups (n < 30) when population SDs are unknown. If we can assume equal population variances, we **pool** the two sample variances into one estimate.

```
H₀: μ₁ = μ₂   Hₐ: μ₁ ≠ μ₂ (or > / <)
s_p² = [(n₁ - 1)s₁² + (n₂ - 1)s₂²] / (n₁ + n₂ - 2)
t = (x̄₁ - x̄₂) / [s_p √(1/n₁ + 1/n₂)]      df = n₁ + n₂ - 2
```

```
Worked: method A n₁ = 10, x̄₁ = 20, s₁ = 4 ; method B n₂ = 12, x̄₂ = 17, s₂ = 5 ; α = 5% two-tailed
s_p² = (9 × 16 + 11 × 25) / 20 = 419 / 20 = 20.95 → s_p = 4.577
SE = 4.577 × √(1/10 + 1/12) = 4.577 × 0.428 = 1.960
t = (20 - 17) / 1.960 = 1.53 ; df = 20 → t crit = 2.086
|1.53| < 2.086 → do not reject H₀: no significant difference
95% CI = 3 ± 2.086 × 1.960 = (-1.09, 7.09) → includes 0 (same conclusion)
```

Check equal variances first with the **F-test** (larger s^2 / smaller s^2)

Unequal variances → **Welch's t**: $t = (\bar{x}_1 - \bar{x}_2)/\sqrt{s_1^2/n_1 + s_2^2/n_2}$ with adjusted (smaller) df

Assumptions: independent random samples, roughly normal populations

TRAP Two independent groups ≠ paired data. Same people measured twice → use the **PAIRED** t-test.

Paired (dependent) t-test: worked ★★★★★

In plain words: when each value in one group is linked to one in the other (same person before/after, matched pairs), test the **differences**. This removes person-to-person variation and is more powerful.

```
D = after - before ; D̄ = ΣD/n ; s_D = √[Σ(D - D̄)²/(n - 1)]
t = D̄ / (s_D/√n)     df = n - 1
Training: 6 employees, scores after - before: 3, 5, -1, 4, 2, 5
D̄ = 18/6 = 3 ; deviations 0, 2, -4, 1, -1, 2 → squares sum = 26
s_D = √(26/5) = 2.280 ; SE = 2.280/√6 = 0.931
t = 3 / 0.931 = 3.22 ; df 5, two-tailed 5% crit = 2.571
3.22 > 2.571 → reject H₀: training changed scores
(one-tailed 'improved?' crit 2.015 → also reject)
```

Which mean test? (summary) ★★★★★

Situation	Test	Statistic	df
One mean, σ known or n ≥ 30	z	$(\bar{x} - \mu)/(\sigma/\sqrt{n})$	—
One mean, σ unknown, n < 30	t	$(\bar{x} - \mu)/(s/\sqrt{n})$	n − 1
Two independent means, large	z	$(\bar{x}_1 - \bar{x}_2)/\sqrt{(s_1^2/n_1 + s_2^2/n_2)}$	—
Two independent means, small, equal σ^2	Pooled t	$(\bar{x}_1 - \bar{x}_2)/[s_p\sqrt{(1/n_1 + 1/n_2)}]$	$n_1 + n_2 - 2$
Two means, same subjects	Paired t	$\bar{D}/(s_D/\sqrt{n})$	n − 1
One proportion	z	$(\hat{p} - p_0)/\sqrt{(p_0q_0/n)}$	—
Two proportions	z	$(\hat{p}_1 - \hat{p}_2)/\sqrt{[\bar{p}\bar{q}(1/n_1 + 1/n_2)]}$, $\bar{p} = (x_1 + x_2)/(n_1 + n_2)$	—
Two variances	F	s_1^2/s_2^2 (larger on top)	$n_1 - 1$, $n_2 - 1$
3+ means	ANOVA F	MSB/MSW	k − 1, N − k

Power of a test ★★★★★

In plain words: power = the chance the test **catches** a real effect = P(reject H_0 when H_0 is false) = 1 − β. Low power means real differences are missed (Type II errors).

```
Worked: H₀: μ = 100 vs Hₐ: μ > 100 ; σ = 15, n = 36 → SE
= 2.5 ; α = 5% (z = 1.645)
Reject H₀ if x̄ > 100 + 1.645 × 2.5 = 104.11
If true μ = 106: β = P(x̄ ≤ 104.11 | μ = 106) = P(z ≤
(104.11 - 106)/2.5) = P(z ≤ -0.755) = 0.225
Power = 1 - 0.225 = 0.775
```

Change	Effect on power
Larger sample n	Up
Larger true difference (effect size)	Up
Higher α (e.g. 10% vs 5%)	Up (but more Type I errors)
Smaller σ (less noise)	Up
One-tailed instead of two-tailed (right direction)	Up

Power curve: plot power against possible true μ values; operating characteristic (OC) curve plots β

Common target: power ≥ 0.80 when planning sample size

TRAP α and β move in opposite directions for a fixed n — only a bigger sample cuts both.

Degrees of freedom ★★★★★

In plain words: degrees of freedom (df) = number of values free to vary after you use the data to estimate something. Knowing the mean of 5 numbers, only 4 can be chosen freely — the last is fixed. df picks the right row in t, χ^2 and F tables.

Test	df
Sample variance / one-sample t	n − 1
Paired t	n − 1 (pairs)
Pooled two-sample t	$n_1 + n_2 - 2$
Chi-square goodness of fit	k − 1 − (parameters estimated)
Chi-square independence	(r − 1)(c − 1)
F-test for two variances	$(n_1 - 1, n_2 - 1)$
One-way ANOVA	between k − 1 ; within N − k ; total N − 1
Two-way ANOVA (interaction)	(r − 1)(c − 1) for interaction

08 Report Writing Mechanics

≈Module 4 · Classes 22–23 marks · 5 subtopics · chapter priority ★★★★★

Purpose and types of research reports ★★★★☆

In plain words: the report is the only thing most managers see. A good study with a poor report is often ignored. Write for the reader: clear, short, focused on the decision.

Type	Use
Technical report	Full method, data, analysis — for specialists
Popular / management report	Findings and recommendations in plain language — for decision makers
Short report (memo, letter)	Small studies, progress updates
Long (formal) report	Major projects
Oral presentation	Explain and discuss findings with the client

Structure and components of a research report ★★★★★

Part	Contents
Prefatory (front) part	
Title page	Title, client, researcher, date
Letter of transmittal	Hands the report to the client; summarises the project
Letter of authorisation	Client's approval to do the research
Table of contents	Sections, list of tables, figures, appendices
Executive summary	1–2 pages: problem, approach, design, key findings, conclusions, recommendations — often the only part read
Main body	
Introduction / problem definition	Background, decision problem, research problem, objectives
Approach to the problem	Theory, models, research questions, hypotheses
Research design	Type of design, information needed, data collection, scaling, questionnaire, sampling
Data analysis	Methods used and why
Results / findings	Organised by objective; tables and charts
Limitations and caveats	Time, budget, sample, non-response, measurement problems
Conclusions and recommendations	Answer the problem; actions for management
End (appended) part	
Appendices	Questionnaire, detailed tables, technical details
Bibliography / references	All sources in one citation style (APA, Harvard)

Some formats also add: abstract, acknowledgements, glossary, index

Guidelines for writing ★★★★★

Guideline	Meaning
Know the readers	Technical depth to suit them; use their language
Easy to follow	Logical structure, headings, short paragraphs
Presentable	Professional layout, page numbers, consistent format
Objective	Report results honestly, including findings against expectations
Reinforce text with visuals	Tables and graphs for key points
Concise	No unnecessary words; but complete
Findings vs conclusions vs recommendations	Findings = facts from data; conclusions = what they mean; recommendations = what to do

TRAP Do not put raw computer output in the main body — put it in the appendix.

Tables and graphs ★★★★☆

Table element	Rule
Title and number	Short, descriptive; Table 1, 2…
Arrangement	Order rows by size or logic
Basis of measurement	Units and base (n, %) shown
Leaders, rulings, spaces	Guide the eye
Headings and stubs	Label columns and rows
Footnotes and sources	Explain and credit

Graph	Best for
Pie chart	Parts of a whole (few categories)
Bar / column chart	Comparing categories
Line chart	Trends over time
Histogram	Distribution of a continuous variable
Pictograph, map	Simple visual messages, regions
Flow chart	Processes and steps

Oral presentation and follow-up ★★★★☆

Know the audience; rehearse; use visuals; tell → show → tell; keep eye contact; leave time for questions (KISS: keep it simple and straightforward)

Help the client understand and act on the report; answer questions after delivery

Evaluate the project afterwards: what to do better next time

Ethics: report honestly, no selective reporting or misuse of statistics, protect respondent confidentiality

09 Coverage Map: Session Plan vs Your Notes

≈— marks · 1 subtopics · chapter priority ★★★★☆

Where to study each session-plan topic ★★★★☆

Session-plan topic	Class	Where
Introduction to research, problem formulation	1	This PDF, Ch 1
Research design, types of research	2	This PDF, Ch 2 (+ Statistics Summary Ch 1)
Primary and secondary data	3–4	This PDF, Ch 3
Sampling techniques	3–4	Statistics Summary Ch 2 (+ this PDF Ch 3)
Measurement scaling techniques	5–6	This PDF, Ch 4 (+ Statistics Summary Ch 1 scales)
Questionnaire designing	5–6	This PDF, Ch 5
Hypotheses, LOS, Type I/II, power, df	7–8	Statistics Summary Ch 6 + this PDF Ch 7
Point and interval estimates; differences	9–10	This PDF, Ch 6
One-sample, two-sample, paired t	11–13	Statistics Summary Ch 7 + this PDF Ch 7
z tests for means and proportions, F-test, ANOVA	14–18	Statistics Summary Ch 7, 8, 10
Chi-square goodness of fit and contingency table	19–21	Statistics Summary Ch 9
Report structure and components	22–23	This PDF, Ch 8